Assignment report

by Kirill Shumskii

Methodology

1. Data preparation

Since given data preparation script didn't work because of lack of RAM on my machine, I needed to implement some modifications. I modified `create_doc` function to upload .txt documents in hdfs storage, also I decreased batch size to 32. In addition I've added document text cleaning.

2. Indexing

This pipeline satisfies minimal functional requirements for implementing inverted index and was first natural choice for full-text search engine. I have decided to index 1000 docs to test performance on larger slice

Map

For each input line (`doc_id`, `title`, `text`):

1. Tokenizes text with regex `[a-z0-9]+`.
2. Computes document length (`doc_len = number of tokens`).
3. Computes term frequency (`tf`) per term in the document.
4. Emits records:

```
term\tdoc_id\ttf\tdoc_len\ttitle
```

Reduce

For each grouped term:

1. Collects all postings for that term.
2. Computes document frequency `df = len(postings)`.
3. Emits one record per posting with attached `df`:

```
term\tdf\tdoc_id\ttf\tdoc_len\ttitle
```

3. Storage design

I have designed 4 tables in cassandra:

- `vocabulary(term -> df)` for fast term metadata lookup.
- `postings((term), doc_id -> tf)` for fast retrieval of all docs containing a term.

- `doc_stats(doc_id -> title, doc_len)` for document metadata and normalization factors.
- `global_stats(id -> total_docs, avg_doc_len)` for collection-level ranking constants.

Since I have precomputed statistics for docs and terms, such denormalization avoids heavy joins and follows query driven design. This design allows efficiently compute all needed terms for BM25 full text search.

4. Querying and ranking

I have designed standard BM25 query pipeline that:

1. Tokenizes query terms using the same regex as indexing
2. Reads all needed info from cassandra (global, postings, docs meta)
3. Computes BM25 per posting in Spark RDD then sums by `doc_id`
4. Sorts results by score

BM25 params: `K1 = 1.2 B = 0.75`

Demonstration

How to run guide

1. Run `docker compose up -d` from the root of the project.
2. Master node will execute `app.sh` automatically. This script runs all of the steps of the assignment.
3. Queries results could be found in `cluster-master` logs, since they are printed in stdout (marked as `Top 10 results for: "query text ..."`)
4. You could run `docker compose logs -f cluster-master > master_logs.txt` for more convenient display.

Reflection

Pipeline successfully worked on distributed system and gave meaningful results for my searches. Execution time for all stages (preparation, indexing, querying) is moderate and expected considering data volume and available hardware.

Examples of queries:

```
============================================================
Top 10 results for: "world war history"
============================================================
1. 55441239 A_History_of_the_Modern_World (score: 9.2885)
2. 73467161
A_History_of_the_Negro_Troops_in_the_War_of_the_Rebellion,_1861–1865
(score: 7.6598)
3. 64470348 A_Cry_from_the_Grave (score: 7.6054)
4. 851866 A_Fistful_of_TOWs (score: 7.1316)
5. 56380311 A_Bear_Named_Winnie (score: 6.9999)
6. 52844930 A_Call_for_Arms (score: 6.7662)
7. 7675887 A_Fortunate_Life (score: 6.7325)
8. 200010 A_Boy_and_His_Dog (score: 6.6741)
9. 13958799 A_Dying_Colonialism (score: 6.6287)
```

```
10. 3740022 A_Gun_for_Sale (score: 6.4794)
==============================================================
```

```
==============================================================
Top 10 results for: "music album rock"
==============================================================
1. 3878521 A_Little_Ain't_Enough (score: 9.2830)
2. 3274533 A_Flight_and_a_Crash (score: 9.2795)
3. 46860526 A_Beautiful_Soul_(song) (score: 9.2782)
4. 54197670 A_Deeper_Understanding (score: 9.2014)
5. 50813860 A_History_of_Garage_and_Frat_Bands_in_Memphis_1960–
1975,_Volume_1 (score: 8.4405)
6. 74860 A_Hard_Day's_Night_(album) (score: 8.2996)
7. 6495360 A_Foreign_Affair_II (score: 8.2913)
8. 38752487 A_Glimmer_of_Hope (score: 8.1474)
9. 6525208 A_Date_with_The_Smithereens (score: 8.1141)
10. 6141063 A_Dios_le_Pido (score: 8.1115)
==============================================================
```

```
==============================================================
Top 10 results for: "united states president"
==============================================================
1. 56881426 A_Day_in_the_Life_of_Marlon_Bundo (score: 9.5660)
2. 15433209 A_Distant_Trumpet (score: 8.4890)
3. 38212997 A_Goon's_Deed_in_a_Weary_World (score: 7.8889)
4. 8605635 A_Christmas_Story_House (score: 7.7717)
5. 7662078 A_Just_Russia_–_For_Truth (score: 7.6843)
6. 10174562 A_History_of_Money_and_Banking_in_the_United_States (score:
6.8292)
7. 16932717 A_Guest_of_Honor_(opera) (score: 6.5479)
8. 58595904 A_Better_Chance (score: 6.3020)
9. 8021609 A_&_M_Karagheusian (score: 6.1901)
10. 4230473 A_History_of_the_American_People (score: 6.1809)
==============================================================
```

Screenshots

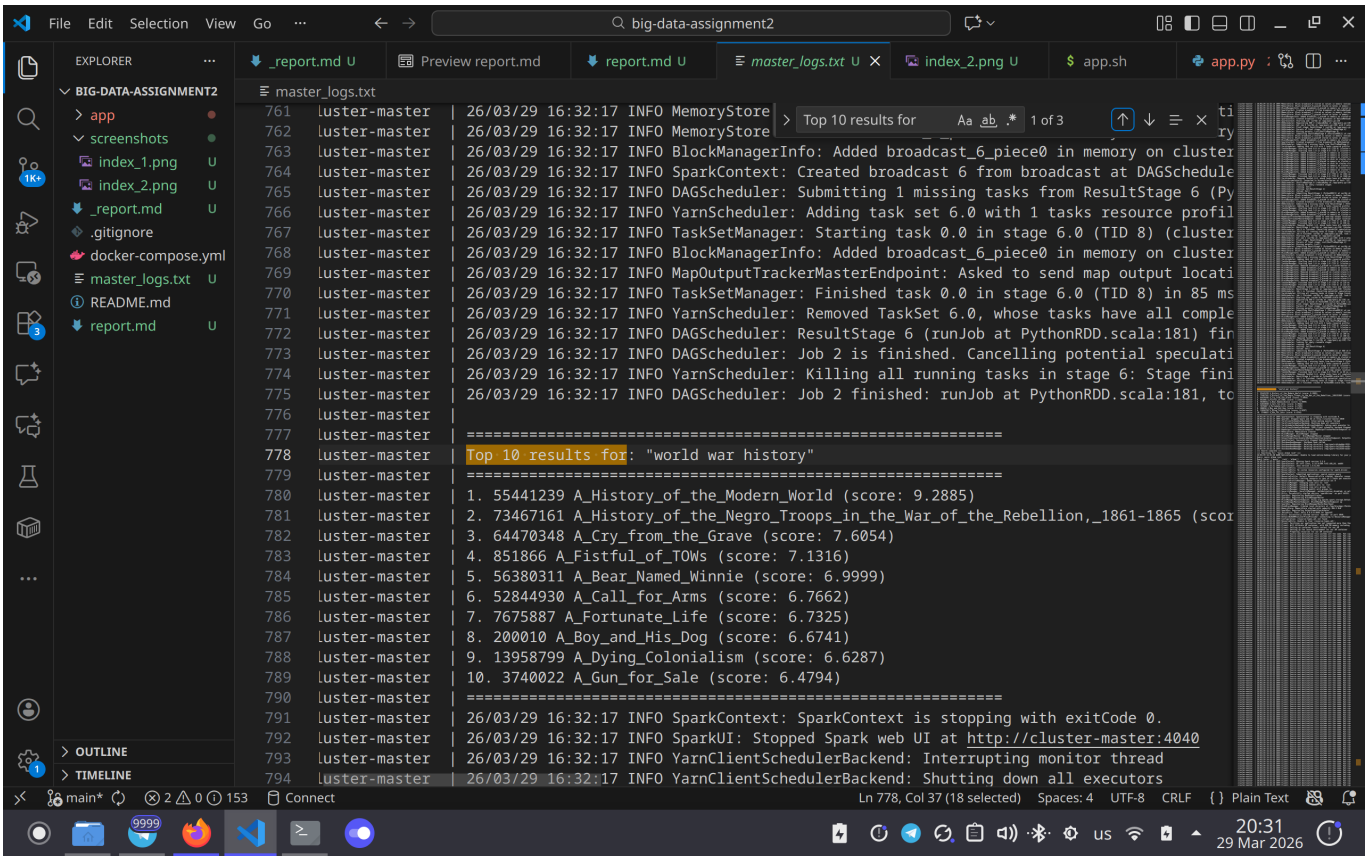
Indexing 1000 docs

```
cluster-master | -rw-r--r--   1 root supergroup    3535569 2026-03-29
cluster-master | === Running indexer ===
cluster-master | === Full indexing pipeline ===
cluster-master | Step 1: Creating index with MapReduce...
cluster-master | === Creating index using MapReduce ===
cluster-master | Input path: /input/data
cluster-master | Output path: /indexer/index
cluster-master | Using streaming jar: /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.1.jar
cluster-master | 2026-03-29 16:31:10,659 WARN streaming.StreamJob: -file option is deprecated, please use generic option
cluster-master | packageJobJar: [mapreduce/mapper1.py, mapreduce/reducer1.py, /tmp/hadoop-unjar1000406855477992 52/] [] /
cluster-master | 2026-03-29 16:31:11,630 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager a
cluster-master | 2026-03-29 16:31:11,934 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager a
cluster-master | 2026-03-29 16:31:12,225 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/had
cluster-master | 2026-03-29 16:31:12,746 INFO mapred.FileInputFormat: Total input files to process : 1
cluster-master | 2026-03-29 16:31:12,861 INFO mapreduce.JobSubmitter: number of splits:2
cluster-master | 2026-03-29 16:31:13,071 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1774801478515_0001
cluster-master | 2026-03-29 16:31:13,072 INFO mapreduce.JobSubmitter: Executing with tokens: []
cluster-master | 2026-03-29 16:31:13,262 INFO conf.Configuration: resource-types.xml not found
cluster-master | 2026-03-29 16:31:13,262 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
cluster-master | 2026-03-29 16:31:13,397 INFO impl.YarnClientImpl: Submitted application application_1774801478515_0001
cluster-master | 2026-03-29 16:31:13,426 INFO mapreduce.Job: The url to track the job: http://cluster-master:8088/proxy/
cluster-master | 2026-03-29 16:31:13,426 INFO mapreduce.Job: Running job: job_1774801478515_0001
cluster-master | 2026-03-29 16:31:20,543 INFO mapreduce.Job: Job job_1774801478515_0001 running in uber mode : false
cluster-master | 2026-03-29 16:31:20,544 INFO mapreduce.Job:  map 0% reduce 0%
cluster-master | 2026-03-29 16:31:25,607 INFO mapreduce.Job:  map 100% reduce 0%
cluster-master | 2026-03-29 16:31:29,646 INFO mapreduce.Job:  map 100% reduce 100%
cluster-master | 2026-03-29 16:31:30,667 INFO mapreduce.Job: Job job_1774801478515_0001 completed successfully
cluster-master | 2026-03-29 16:31:30,751 INFO mapreduce.Job: Counters: 54
cluster-master |   File System Counters
cluster-master |       FILE: Number of bytes read=11895301
cluster-master |       FILE: Number of bytes written=24620062
cluster-master |       FILE: Number of read operations=0
cluster-master |       FILE: Number of large read operations=0
cluster-master |       FILE: Number of write operations=0
```

```
cluster-master |       WRONG_MAP=0
cluster-master |       WRONG_REDUCE=0
cluster-master |   File Input Format Counters
cluster-master |       Bytes Read=3545449
cluster-master |   File Output Format Counters
cluster-master |       Bytes Written=12156545
cluster-master | 2026-03-29 16:31:30,751 INFO streaming.StreamJob: Output directory: /indexer/index
cluster-master | MapReduce job completed
cluster-master | Found 2 items
cluster-master | -rw-r--r--   1 root supergroup          0 2026-03-29 16:31 /indexer/index/_SUCCESS
cluster-master | -rw-r--r--   1 root supergroup   12156545 2026-03-29 16:31 /indexer/index/part-00000
cluster-master | === Index creation done ===
cluster-master | Step 2: Storing index in Cassandra...
cluster-master | === Storing index data to Cassandra ===
cluster-master | Reading index from HDFS: /indexer/index
cluster-master | Found 2 items
cluster-master | -rw-r--r--   1 root supergroup          0 2026-03-29 16:31 /indexer/index/_SUCCESS
cluster-master | -rw-r--r--   1 root supergroup   12156545 2026-03-29 16:31 /indexer/index/part-00000
cluster-master | Loading data into Cassandra...
cluster-master | Connected to Cassandra on attempt 1
cluster-master | Cassandra schema created
cluster-master | Loaded 40218 vocabulary terms
cluster-master | Loaded 251030 postings
cluster-master | Loaded 1000 document stats
cluster-master | Average document length: 580.82
cluster-master | Index data stored in Cassandra successfully
cluster-master | === Index stored in Cassandra ===
cluster-master | === Full indexing pipeline completed ===
cluster-master | === Running sample searches ===
cluster-master | === Searching for: "world war history" ===
cluster-master | 26/03/29 16:31:50 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... usin
cluster-master | Query: world war history
cluster-master | Query terms: ['history', 'world', 'war']
```

Queries

Q1


master_logs.txt
luster-master | 26/03/29 16:32:17 INFO MemoryStore
luster-master | 26/03/29 16:32:17 INFO BlockManagerInfo: Added broadcast_6_piece0 in memory on cluster
luster-master | 26/03/29 16:32:17 INFO SparkContext: Created broadcast 6 from broadcast at DAGSchedule
luster-master | 26/03/29 16:32:17 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 6 (Py
luster-master | 26/03/29 16:32:17 INFO YarnScheduler: Adding task set 6.0 with 1 tasks resource profil
luster-master | 26/03/29 16:32:17 INFO TaskSetManager: Starting task 0.0 in stage 6.0 (TID 8) (cluster
luster-master | 26/03/29 16:32:17 INFO BlockManagerInfo: Added broadcast_6_piece0 in memory on cluster
luster-master | 26/03/29 16:32:17 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locati
luster-master | 26/03/29 16:32:17 INFO TaskSetManager: Finished task 0.0 in stage 6.0 (TID 8) in 85 ms
luster-master | 26/03/29 16:32:17 INFO YarnScheduler: Removed TaskSet 6.0, whose tasks have all comple
luster-master | 26/03/29 16:32:17 INFO DAGScheduler: ResultStage 6 (runJob at PythonRDD.scala:181) fin
luster-master | 26/03/29 16:32:17 INFO DAGScheduler: Job 2 is finished. Cancelling potential speculati
luster-master | 26/03/29 16:32:17 INFO YarnScheduler: Killing all running tasks in stage 6: Stage fini
luster-master | 26/03/29 16:32:17 INFO DAGScheduler: Job 2 finished: runJob at PythonRDD.scala:181, to
luster-master |
luster-master | ==========
luster-master | Top 10 results for: "world war history"
luster-master | ==========
luster-master | 1. 55441239 A_History_of_the_Modern_World (score: 9.2885)
luster-master | 2. 73467161 A_History_of_the_Negro_Troops_in_the_War_of_the_Rebellion,_1861-1865 (scor
luster-master | 3. 64470348 A_Cry_from_the_Grave (score: 7.6054)
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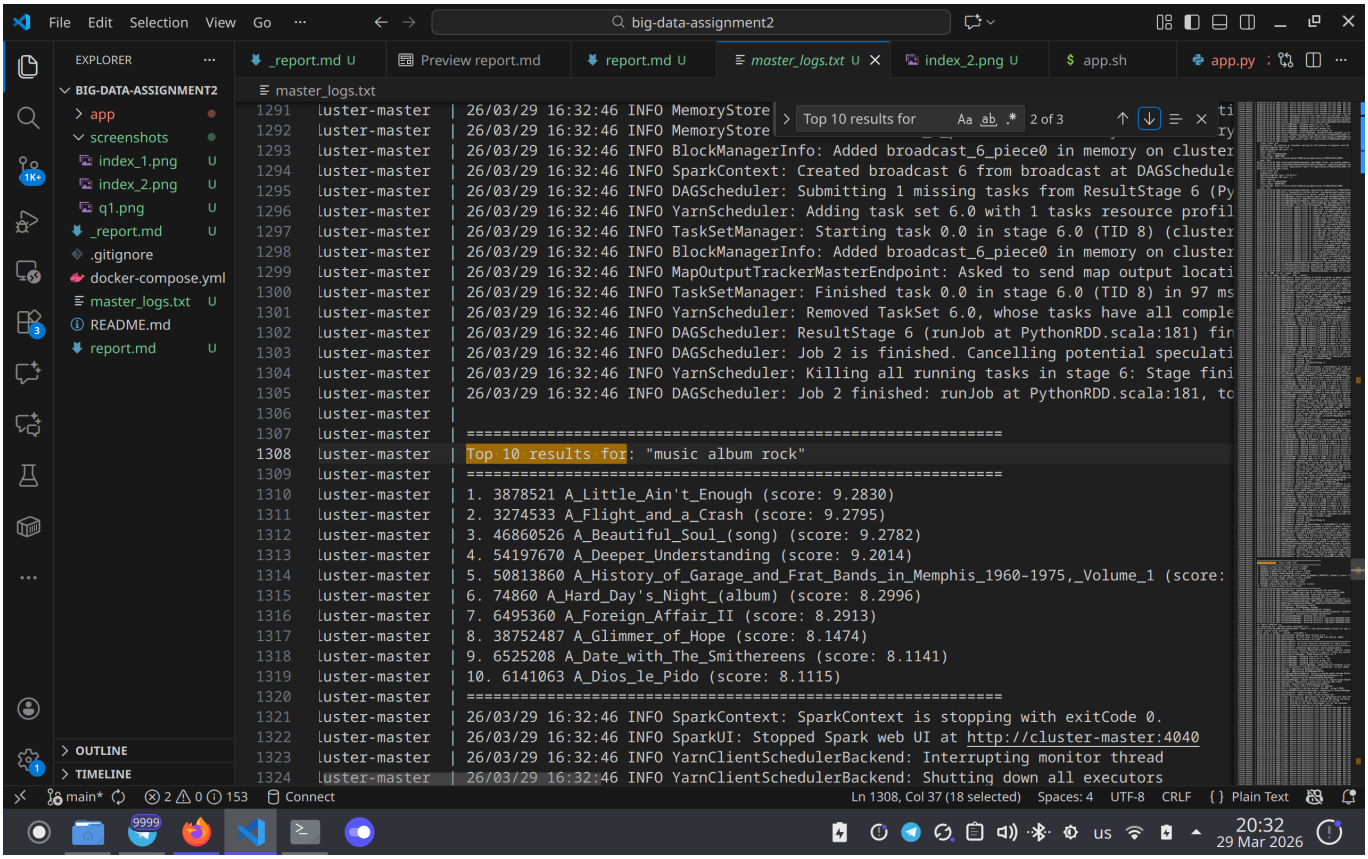
luster-master | 8. 200010 A_Boy_and_His_Dog (score: 6.7041)
luster-master | 9. 13958799 A_Dying_Colonialism (score: 6.6287)
luster-master | 10. 3740022 A_Gun_for_Sale (score: 6.4794)
luster-master | ==========
luster-master | 26/03/29 16:32:17 INFO SparkContext: SparkContext is stopping with exitCode 0.
luster-master | 26/03/29 16:32:17 INFO SparkUI: Stopped Spark web UI at http://cluster-master:4040
luster-master | 26/03/29 16:32:17 INFO YarnClientSchedulerBackend: Interrupting monitor thread
luster-master | 26/03/29 16:32:17 INFO YarnClientSchedulerBackend: Shutting down all executors

Q2


master_logs.txt
luster-master | 26/03/29 16:32:46 INFO MemoryStore
luster-master | 26/03/29 16:32:46 INFO BlockManagerInfo: Added broadcast_6_piece0 in memory on cluster
luster-master | 26/03/29 16:32:46 INFO SparkContext: Created broadcast 6 from broadcast at DAGSchedule
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luster-master | 26/03/29 16:32:46 INFO BlockManagerInfo: Added broadcast_6_piece0 in memory on cluster
luster-master | 26/03/29 16:32:46 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locati
luster-master | 26/03/29 16:32:46 INFO TaskSetManager: Finished task 0.0 in stage 6.0 (TID 8) in 97 ms
luster-master | 26/03/29 16:32:46 INFO YarnScheduler: Removed TaskSet 6.0, whose tasks have all comple
luster-master | 26/03/29 16:32:46 INFO DAGScheduler: ResultStage 6 (runJob at PythonRDD.scala:181) fin
luster-master | 26/03/29 16:32:46 INFO DAGScheduler: Job 2 is finished. Cancelling potential speculati
luster-master | 26/03/29 16:32:46 INFO YarnScheduler: Killing all running tasks in stage 6: Stage fini
luster-master | 26/03/29 16:32:46 INFO DAGScheduler: Job 2 finished: runJob at PythonRDD.scala:181, to
luster-master |
luster-master | ==========
luster-master | Top 10 results for: "music album rock"
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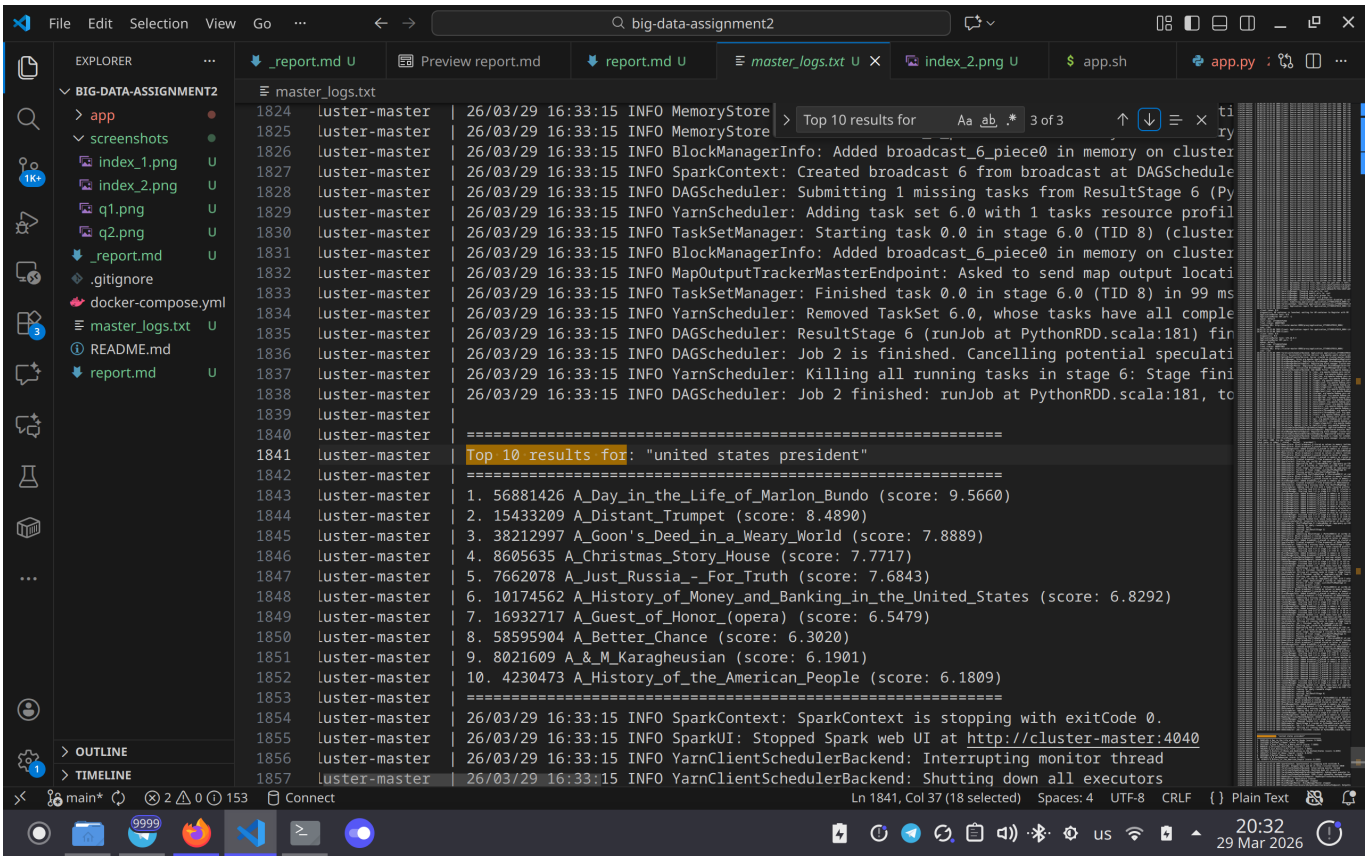
luster-master | 3. 46860526 A_Beautiful_Soul_(song) (score: 9.2782)
luster-master | 4. 54197670 A_Deeper_Understanding (score: 9.2014)
luster-master | 5. 50813860 A_History_of_Garage_and_Frat_Bands_in_Memphis_1960-1975,_Volume_1 (score:
luster-master | 6. 74860 A_Hard_Day's_Night_(album) (score: 8.2996)
luster-master | 7. 6495360 A_Foreign_Affair_II (score: 8.2913)
luster-master | 8. 38752487 A_Glimmer_of_Hope (score: 8.1474)
luster-master | 9. 6525208 A_Date_with_The_Smithereens (score: 8.1141)
luster-master | 10. 6141063 A_Dios_le_Pido (score: 8.1115)
luster-master | ==========
luster-master | 26/03/29 16:32:46 INFO SparkContext: SparkContext is stopping with exitCode 0.
luster-master | 26/03/29 16:32:46 INFO SparkUI: Stopped Spark web UI at http://cluster-master:4040
luster-master | 26/03/29 16:32:46 INFO YarnClientSchedulerBackend: Interrupting monitor thread
luster-master | 26/03/29 16:32:46 INFO YarnClientSchedulerBackend: Shutting down all executors

Q3


File Edit Selection View Go
big-data-assignment2
EXPLORER
BIG-DATA-ASSIGNMENT2
app
screenshots
index_1.png
index_2.png
q1.png
q2.png
_report.md
.gitignore
docker-compose.yml
master_logs.txt
README.md
report.md
OUTLINE
TIMELINE
_report.md U
Preview report.md
report.md U
master_logs.txt U
index_2.png U
app.sh
app.py
master_logs.txt
Top 10 results for
3 of 3
luster-master | 26/03/29 16:33:15 INFO MemoryStore
luster-master | 26/03/29 16:33:15 INFO MemoryStore
luster-master | 26/03/29 16:33:15 INFO BlockManagerInfo: Added broadcast_6_piece0 in memory on cluster
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luster-master | 26/03/29 16:33:15 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locati
luster-master | 26/03/29 16:33:15 INFO TaskSetManager: Finished task 0.0 in stage 6.0 (TID 8) in 99 ms
luster-master | 26/03/29 16:33:15 INFO YarnScheduler: Removed TaskSet 6.0, whose tasks have all comple
luster-master | 26/03/29 16:33:15 INFO DAGScheduler: ResultStage 6 (runJob at PythonRDD.scala:181) fin
luster-master | 26/03/29 16:33:15 INFO DAGScheduler: Job 2 is finished. Cancelling potential speculati
luster-master | 26/03/29 16:33:15 INFO YarnScheduler: Killing all running tasks in stage 6: Stage fini
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luster-master | 4. 8605635 A_Christmas_Story_House (score: 7.7717)
luster-master | 5. 7662078 A_Just_Russia_-_For_Truth (score: 7.6843)
luster-master | 6. 10174562 A_History_of_Money_and_Banking_in_the_United_States (score: 6.8292)
luster-master | 7. 16932717 A_Guest_of_Honor_(opera) (score: 6.5479)
luster-master | 8. 58595904 A_Better_Chance (score: 6.3020)
luster-master | 9. 8021609 A_&_M_Karagheusian (score: 6.1901)
luster-master | 10. 4230473 A_History_of_the_American_People (score: 6.1809)
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luster-master | 26/03/29 16:33:15 INFO YarnClientSchedulerBackend: Interrupting monitor thread
luster-master | 26/03/29 16:33:15 INFO YarnClientSchedulerBackend: Shutting down all executors
main*
Connect
Ln 1841, Col 37 (18 selected)
Spaces: 4
UTF-8
CRLF
Plain Text
20:32
29 Mar 2026